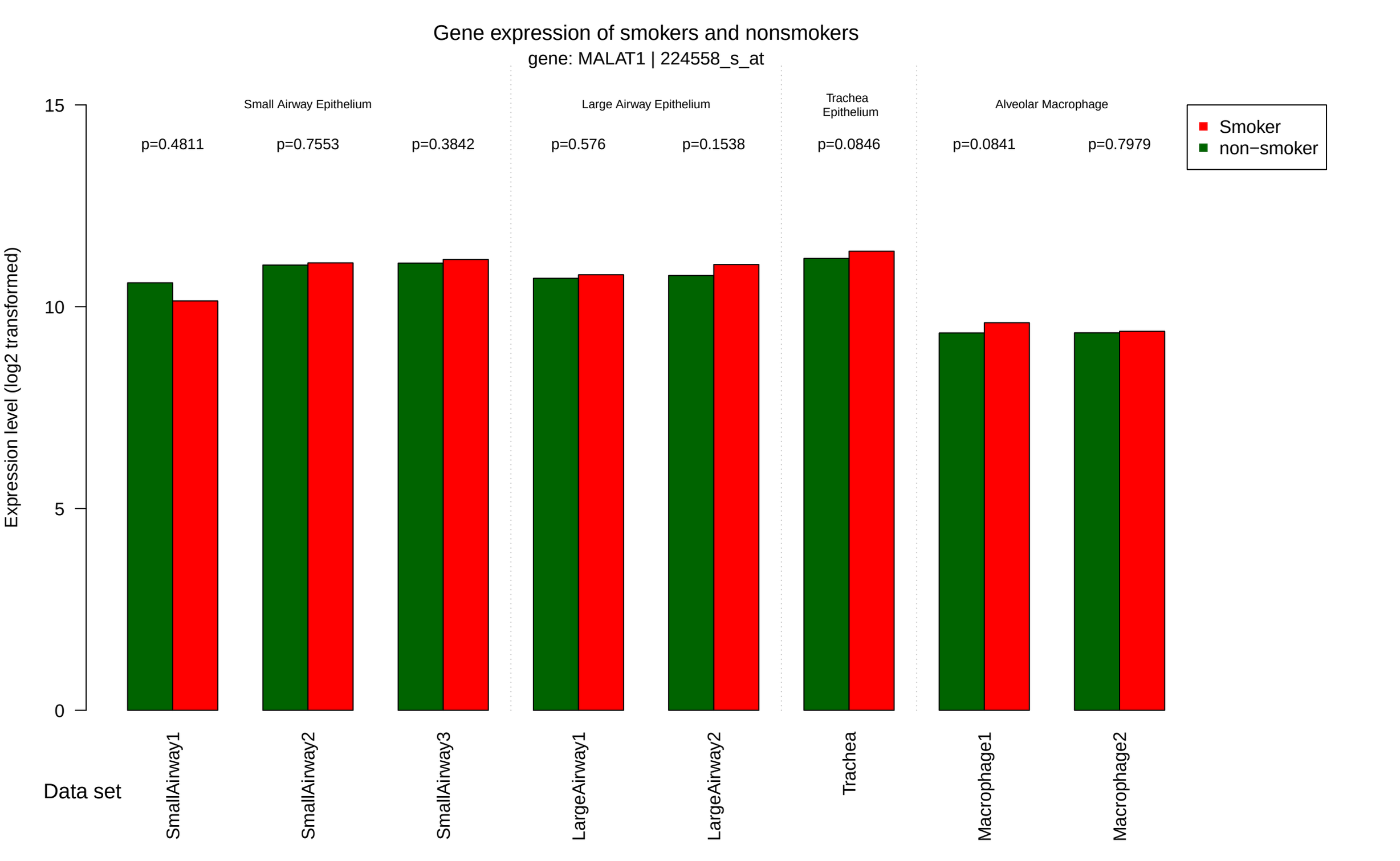
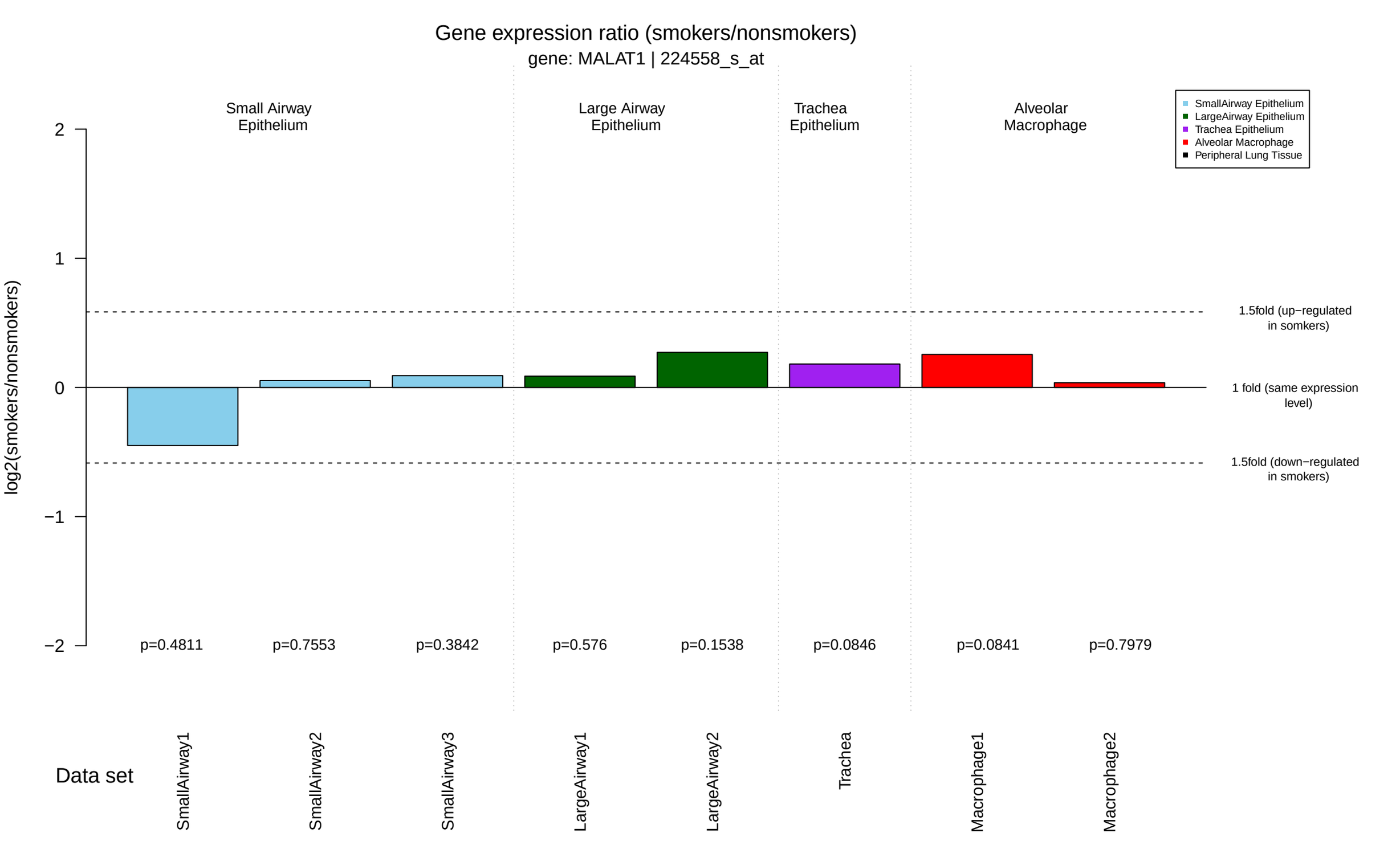
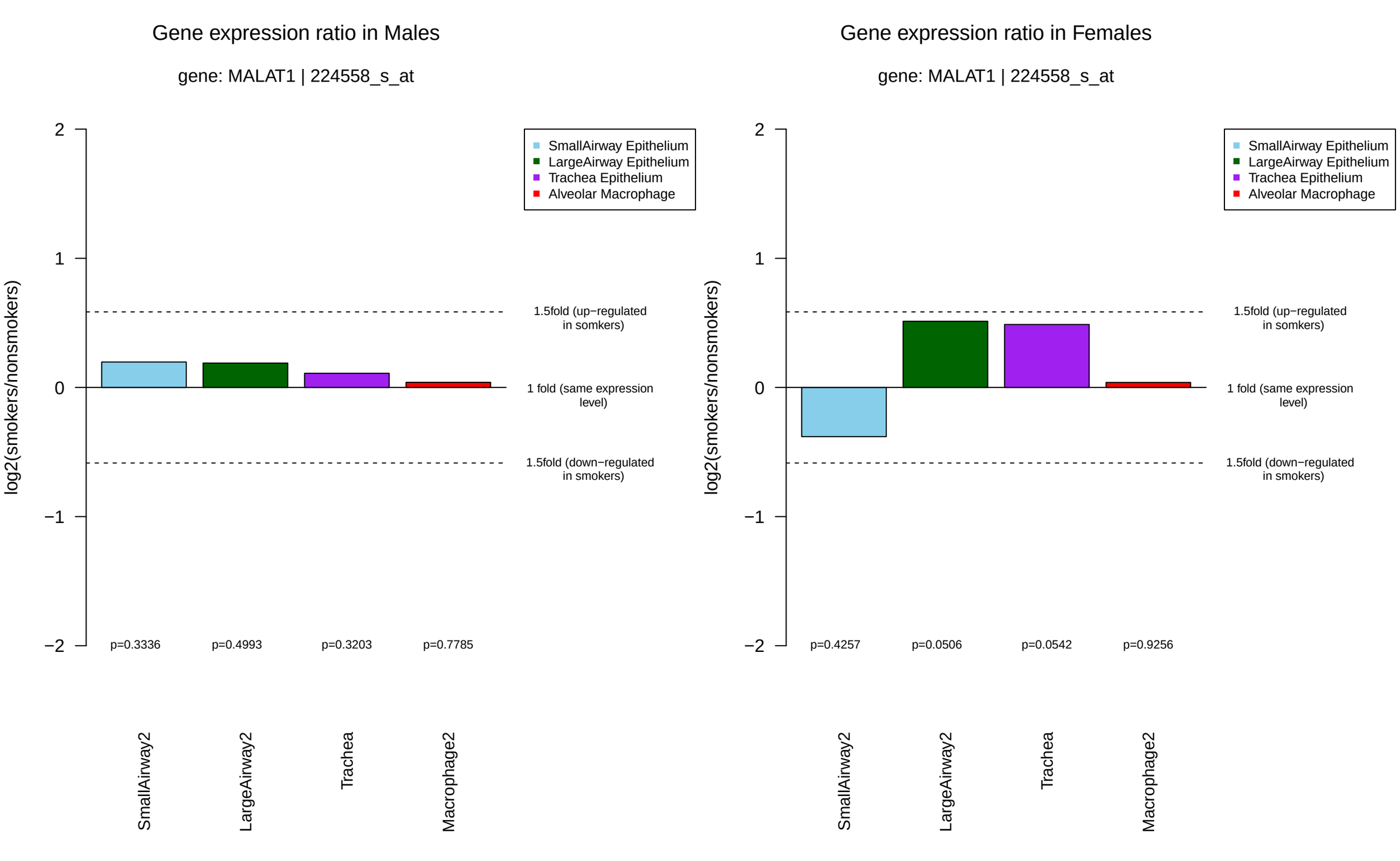
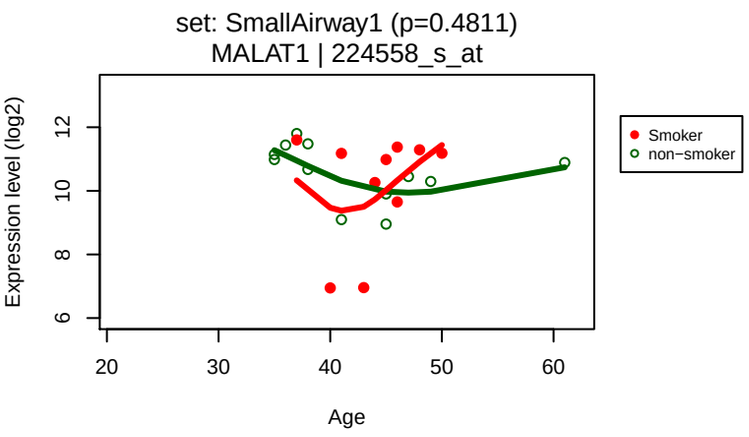
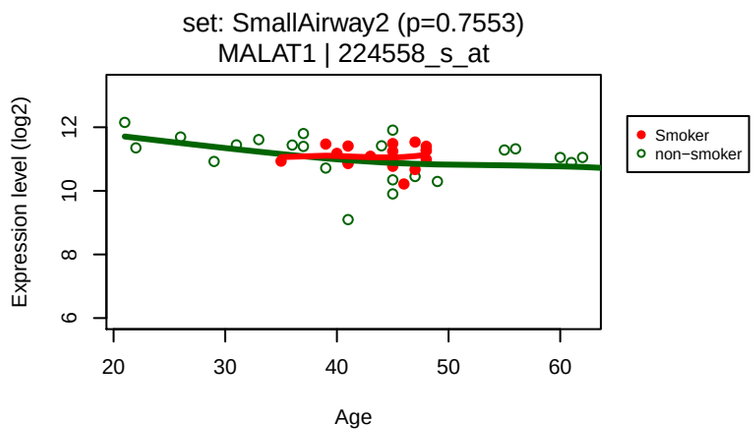
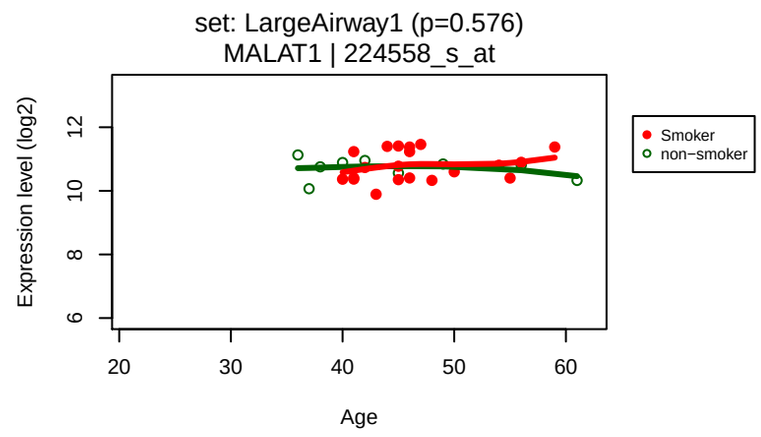

Gene expression of smokers and nonsmokers
gene: MALAT1 | 224558_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4811
p=0.7553
p=0.3842
p=0.576
p=0.1538
p=0.0846
p=0.0841
p=0.7979
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

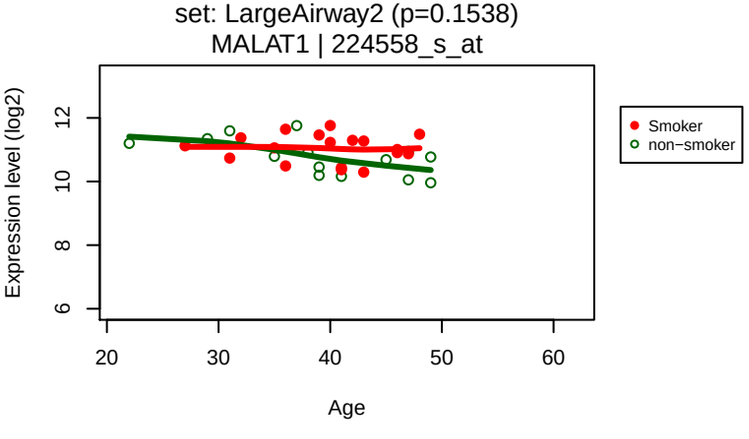

Gene expression ratio (smokers/nonsmokers)
gene: MALAT1 | 224558_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4811
p=0.7553
p=0.3842
p=0.576
p=0.1538
p=0.0846
p=0.0841
p=0.7979
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

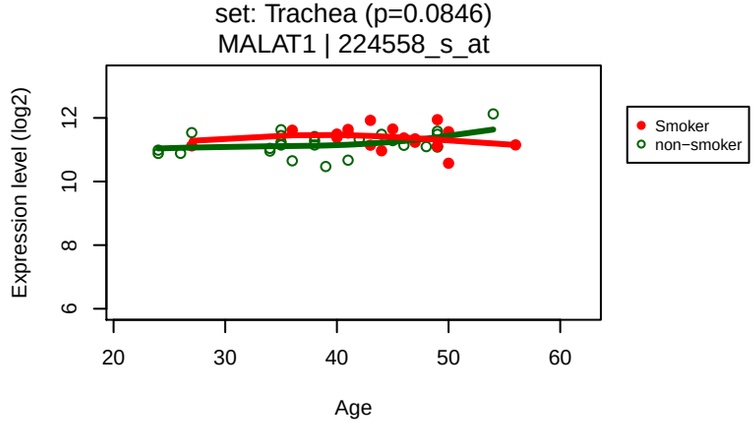

Gene expression ratio in Males
gene: MALAT1 | 224558_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3336
p=0.4993
p=0.3203
p=0.7785
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MALAT1 | 224558_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4257
p=0.0506
p=0.0542
p=0.9256
SmallAirway2
LargeAirway2
Trachea
Macrophage2

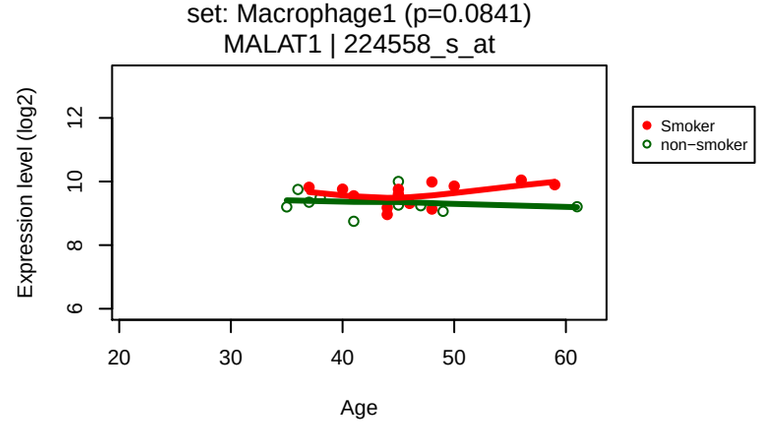

set: SmallAirway1 (p=0.4811)
MALAT1 | 224558_s_at
Expression level (log2)
Age
Smoker
non−smoker

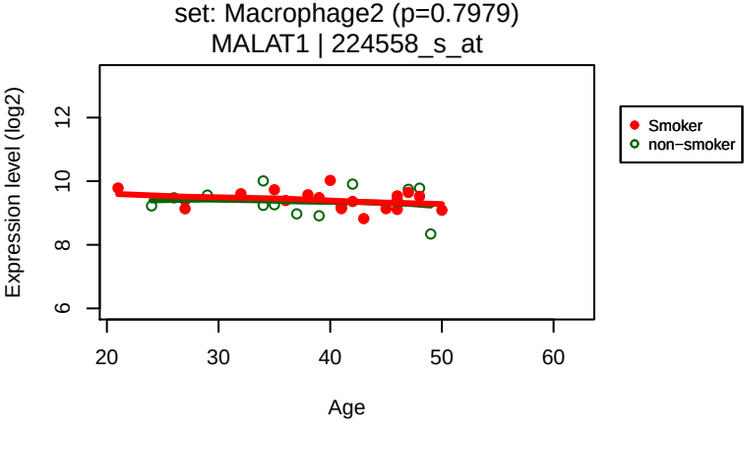

set: SmallAirway2 (p=0.7553)
MALAT1 | 224558_s_at
Expression level (log2)
Age
Smoker
non−smoker

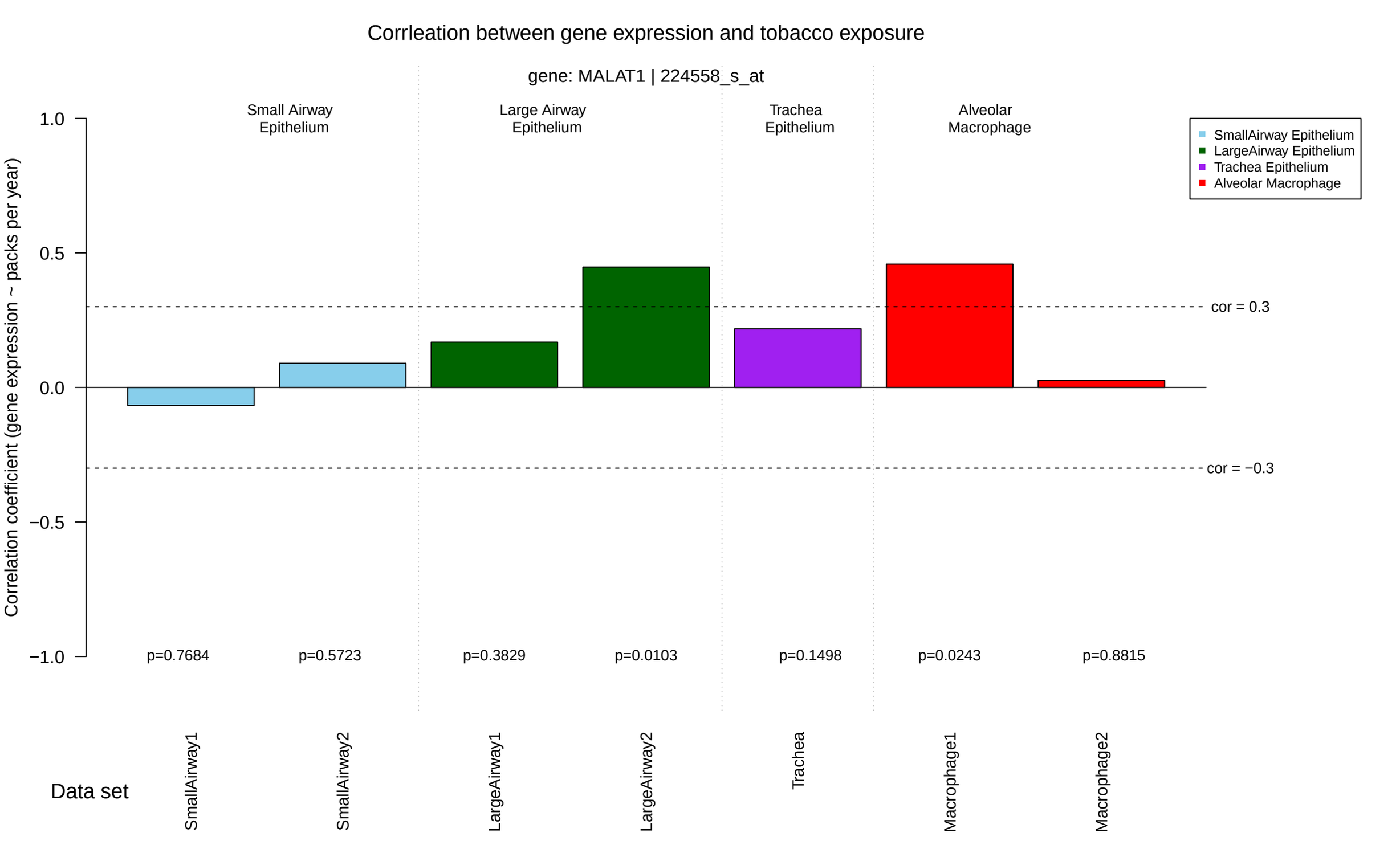

set: LargeAirway1 (p=0.576)
MALAT1 | 224558_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1538)
MALAT1 | 224558_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0846)
MALAT1 | 224558_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0841)
MALAT1 | 224558_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7979)
MALAT1 | 224558_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MALAT1 | 224558_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7684
p=0.5723
p=0.3829
p=0.0103
p=0.1498
p=0.0243
p=0.8815
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2